A four dimensional finite volume criterion for a transfer gap

Research note and proof audit

October 9, 2026

Abstract

We extend the elementary two-direction partition-function argument in the repository's $O(4)$ manuscript to four directions. For a consistently normalized rectangular-torus model admitting positive transfer traces in every coordinate direction, define $D(n) = 16 \log Z(n,n,n,n) - \log Z(2n,2n,2n,2n)$. We prove that $D(n) \le 1/2048$ implies a transfer eigenvalue ratio at most $2^{-1/n}$ on every spatial cube of side $2^k n$. Under explicit local-observable and reflection-positivity hypotheses, the same bound holds in every local limit along these dyadic tori. The argument does not prove that the small-defect test holds on a weak-bare-coupling continuum trajectory. It does not construct a continuum Yang–Mills theory, prove nontriviality, or establish uniqueness of all thermodynamic states. No novelty claim is made for the criterion.

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1 Scope and structural hypotheses

This note concerns the finite-volume criterion of Family 215 in the supplied repository, not its model-specific mass-generation and renormalization estimates. The two-dimensional source is [1, section "A finite-volume criterion for the full gap"]; its use in the continuum argument appears in [2, section "Trace estimates uniform in the cutoff"]. The repository revision inspected is `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`.

For an even side-length vector $x = (x_1, x_2, x_3, x_4)$ with all $x_i \ge 4$, let $Z(x) > 0$ be a torus partition function and $F(x) = \log Z(x)$. Let $D_i x$ double only coordinate i and let $Dx = D_1 D_2 D_3 D_4 x$. These operations commute. Write $\underline{r} = (r, r, r, r)$.

Assumption 1.1 (Positive rectangular transfer traces)**.** For every direction i and fixed cross-section $\widehat{x}_i$ there is a nonzero positive, self-adjoint, compact, trace-class operator $K_{i,\widehat{x}_i}$ such that

$$Z(x) = \operatorname{Tr} K_{i,\widehat{x}_i}^{x_i}$$

for every admitted x_i. Its largest eigenvalue is positive and simple. The same function Z and the same normalization are used in every direction.

Positive means positive semidefinite as an operator, not merely that an integral kernel has positive entries. Only this assumption is needed for the finite-cross-section theorem. Section 5 states the additional hypotheses for local limits.

Define directional and simultaneous doubling defects by

$$p_i(x) = 2F(x) - F(D_i x), \tag{1}$$

$$D(r) = 16F(\underline{r}) - F(2\underline{r}). \tag{2}$$

Multiplying $Z(x)$ by $\exp(c\prod_i x_i)$ leaves every defect unchanged. An arbitrary size-dependent or topology-dependent renormalization need not do so, and need not preserve the transfer-trace hypothesis.

Theorem 1.2 (Four dimensional finite-volume criterion)**.** *Suppose Assumption 1.1 holds. Let $n \geq 4$ be even. If*

$$D(n) \leq \varepsilon_* := 1/2048,$$

then for every $k \geq 0$, every coordinate direction, and the spatial cubic cross-section of side $w_k = 2^k n$, the first two transfer eigenvalues obey

$$\lambda_2(w_k)/\lambda_1(w_k) \leq 2^{-1/n}.$$

If the operator has rank one, read $\lambda_2 = 0$. The full normalized transfer contraction on the orthogonal complement of its leading eigenvector has the same upper bound.

The assertion concerns dyadic cubic cross-sections, not all rectangles or uniqueness of a Gibbs state. A lower bound on $-\log(\lambda_2/\lambda_1)$ does not assert that it is finite.

2 The one-direction spectral estimate

Lemma 2.1 (Trace concentration and squaring)**.** *Let K satisfy the operator hypotheses above, with eigenvalues $\lambda_1 > \lambda_2 \geq \lambda_3 \geq \cdots \geq 0$. Put*

$$p_m = 2\log \operatorname{Tr} K^m - \log \operatorname{Tr} K^{2m}, \qquad s_m = \sum_{j\geq 2} (\lambda_j/\lambda_1)^m.$$

Then $p_m \geq 0$ and $s_m \leq e^{p_m} - 1$. If $p_m \leq 1/4$, then

$$(\lambda_2/\lambda_1)^m \leq 2p_m, \qquad p_{2m} \leq 4p_m^2.$$

In particular $p_i(x) \geq 0$ and

$$p_i(D_i x) \leq 4p_i(x)^2 \quad \text{if } p_i(x) \leq 1/4. \tag{3}$$

Proof. Set $t_j = \lambda_j^m / \operatorname{Tr} K^m$. Then $\sum_j t_j = 1$, $t_1 = (1+s_m)^{-1}$, and

$$e^{-p_m} = \sum_j t_j^2 \le t_1 \sum_j t_j = t_1 \le 1.$$

For $0 \le p_m \le 1/4$, $e^{p_m} - 1 \le p_m e^{p_m} \le 2p_m$. Also $s_{2m} \le s_m^2$, so cancellation of the leading eigenvalue gives

$$p_{2m} = 2\log(1+s_{2m}) - \log(1+s_{4m}) \le 2s_{2m} \le 2p_m^2 e^{2p_m} \le 4p_m^2.$$

□

3 Four-direction algebra

Lemma 3.1 (Telescoping and rectangle identities)**.** *For every ordering $i_1, \dots, i_4$ of the directions,*

$$16F(x) - F(Dx) = \sum_{j=1}^{4} 2^{4-j} p_{i_j}(D_{i_{j-1}} \cdots D_{i_1} x). \tag{4}$$

For distinct i, j,

$$p_i(D_j x) = 2p_i(x) - 2p_j(x) + p_j(D_i x) \le 2p_i(x) + p_j(D_i x). \tag{5}$$

Hence $D(r) \ge 0$, and

$$p_i(x^{[i]}) \le D(r), \qquad x_i^{[i]} = r, \quad x_j^{[i]} = 2r \ (j \ne i). \tag{6}$$

Proof. Expand both equalities; intermediate values of F cancel. The inequality uses $p_j \ge 0$. For (6), telescope from $\underline{r}$ with direction i last. The last term is $p_i(x^{[i]})$ with coefficient one, and the other terms are nonnegative. □

Fix the order $1, 2, 3, 4$ and write $\delta = D(r)$. Define

$$y_j^{(i)} = \begin{cases} 4r & j < i, \\ 2r & j \ge i, \end{cases} \qquad u_i = p_i(y^{(i)}),$$

$$z_j^{(i)} = \begin{cases} 4r & j < i, \\ r & j = i, \\ 2r & j > i, \end{cases} \qquad a_i = p_i(z^{(i)}).$$

Thus $u_i \le 4a_i^2$ whenever $a_i \le 1/4$, and

$$D(2r) = 8u_1 + 4u_2 + 2u_3 + u_4. \tag{7}$$

Lemma 3.2 (Preparing a coordinate before squaring)**.** *For each $i = 1, 2, 3, 4$,*

$$a_i \le 2^{i-1}\delta + \sum_{j<i} 2^{i-1-j} u_j. \tag{8}$$

Proof. Start at $x^{[i]}$, where $p_i \le \delta$. Double the coordinates $j = 1, \dots, i-1$ in order. At the step for j, the box x has earlier coordinates $4r$, coordinate i equal to r, and the others equal to $2r$. Thus $D_i x = y^{(j)}$, and the last term in (5) is exactly u_j. Each step replaces a previous bound A by $2A + u_j$. Iterating proves (8). No unproved monotonicity in transverse size is being used. □

Proposition 3.3 (Quadratic improvement)**.** *If $D(r) \le 1/2048$, then*

$$D(2r) \le 700 D(r)^2 \le 1024 D(r)^2. \tag{9}$$

Proof. Use Lemmas 3.2 and 2.1 successively:

i	unsquared bound	a_i bounded by	u_i bounded by
1	δ	δ	$4\delta^2$
2	$2\delta + 4\delta^2$	3δ	$36\delta^2$
3	$4\delta + 44\delta^2$	5δ	$100\delta^2$
4	$8\delta + 188\delta^2$	9δ	$324\delta^2$

Indeed $4\delta, 44\delta, 188\delta \le 1$ and $9\delta \le 9/2048 < 1/4$, so all squaring steps are legitimate. Equation (7) now gives

$$D(2r) \le (8 \cdot 4 + 4 \cdot 36 + 2 \cdot 100 + 324)\delta^2 = 700\delta^2.$$

□

4 Iteration and the finite-cross-section gap

Proof of Theorem 1.2. Set $C = 1024$, $\delta_k = D(2^k n)$, and $q = C\delta_0 \le 1/2$. Equation (9) preserves the threshold. Induction gives

$$\delta_k \le C^{-1} q^{2^k} \le C^{-1} 2^{-2^k}. \tag{10}$$

At the cube $\underline{w_k}$, telescope with the chosen time direction first, so $8p_i(\underline{w_k}) \le \delta_k$. Lemma 2.1 implies

$$(\lambda_2(w_k)/\lambda_1(w_k))^{w_k} \le 2p_i(\underline{w_k}) \le \delta_k/4 \le (4C)^{-1} 2^{-2^k}.$$

Taking the $w_k = 2^k n$ root proves the claim. □

The proof includes $D(n) = 0$: the excited eigenvalue ratio is then zero. This does not assert the existence of nontrivial finite-energy excitations.

5 Local limits and the full observable Hilbert space

Assumption 5.1 (Local-observable interface)**.** The partition functions come from Borel probability laws on products of compact metrizable local configuration spaces, invariant under translations and the relevant time reflections. Normalize $\widehat{K}_w = K_w/\lambda_1(w)$ and its leading vector Ω_w. Every bounded local observable A in a time slab of thickness r has an insertion $K_{w,A}$ satisfying

$$A_w := K_{w,A}/\lambda_1(w)^r, \qquad \|A_w\| \le \|A\|_\infty ,$$

For torus periods $w > r$ containing the support, the torus and infinite-time cylinder expectations are

$$\omega_{w,\mathrm{tor}}(A) = \frac{\mathrm{Tr}(A_w \widehat{K}_w^{w-r})}{\mathrm{Tr}\, \widehat{K}_w^w}, \qquad \omega_{w,\mathrm{cyl}}(A) = \langle \Omega_w, A_w \Omega_w \rangle.$$

Separated insertions compose through powers of $\widehat{K}_w$. The limiting states retain these invariances. On their admitted positive-time local algebra, let Θ denote time reflection followed by complex conjugation and set $\langle [A], [B] \rangle = \omega((\Theta A)B)$. This form is positive semidefinite. The Hilbert space

is its quotient completion, with $\Omega = [1]$. The *canonical one-step translation* $T[A] = [\tau_1 A]$ is well-defined on this quotient and extends to a positive self-adjoint contraction. In particular $T\Omega = \Omega$ and $\langle [A], T^s[B]\rangle = \omega((\Theta A)\tau_s B)$. For a gauge theory the admitted algebra is gauge-invariant, the marginal laws are gauge invariant, and the compact gauge group acts continuously; the completion is the physical local-observable vacuum representation.

For a positive integral kernel, $|K_{w,A}| \le \|A\|_\infty K_w^r$ implies the insertion norm: bound a matrix element using $|f|, |g|$ and $\|K_w^r\| = \lambda_1(w)^r$. The insertion need not itself be positive.

Proposition 5.2 (Cylinder clustering and torus comparison)**.** *Under Assumptions 1.1 and 5.1 and the small-test hypothesis, put $\rho = 2^{-1/n}$. Bounded observables A, B separated by s transfer steps satisfy*

$$|\operatorname{Cov}_{\omega_{w_k,\mathrm{cyl}}}(A, B)| \le \|A\|_\infty \|B\|_\infty \rho^s. \tag{11}$$

Every local subsequential limit of the cubic tori of side w_k obeys the same inequality. Its full physical transfer operator satisfies $\|T|_{\Omega^\perp}\| \le \rho$.

Proof. Insert $\widehat{K}_{w_k}^s - P_{\Omega_{w_k}}$ between A_{w_k} and B_{w_k} and apply Theorem 1.2 to obtain (11). First take continuous local observables. For a fixed local A, take $k \ge 1$ large enough that $r \le w_k/2$. Set $s_m(w) = \operatorname{Tr} \widehat{K}_w^m - 1$. Separating the leading eigenvector in numerator and denominator shows

$$|\omega_{w_k,\mathrm{tor}}(A) - \omega_{w_k,\mathrm{cyl}}(A)| \le 2\|A\|_\infty s_{w_k/2}(w_k). \tag{12}$$

The opposite-face telescoping identity gives

$$p_1(w_k/2, w_k, w_k, w_k) \le D(w_k/2) = \delta_{k-1}.$$

Thus $s_{w_k/2}(w_k) \le e^{\delta_{k-1}} - 1 \to 0$. Compactness supplies local weak subsequential limits; the comparison identifies torus and cylinder limits along the same subsequence. Apply it to continuous A, B, and AB, with fixed supports, to pass (11) to the limit. For bounded measurable local observables, work in the limiting finite-support marginals: approximate in L^2 by continuous functions with the same supremum bounds. Gauge averaging preserves continuity, gauge invariance and these bounds. Cauchy–Schwarz then passes the covariance bound to the measurable observables. This is not a claim that weak convergence itself implies convergence of every bounded measurable expectation.

A centered vector v_A from a local positive-time observable has $\langle v_A, T^s v_A\rangle$ equal to a covariance of reflected slabs. Their finite thickness changes only a prefactor, so

$$0 \le \langle v_A, T^s v_A\rangle \le C_A \rho^s$$

for all sufficiently large integer s. Finite linear combinations retain the same ρ. A positive spectral measure with nonzero mass in $[b, 1]$ for some $b > \rho$ would give a lower bound proportional to b^s, a contradiction. Bounded spectral projections and density extend the exclusion to all of $\Omega^\perp$. □

This concerns each selected local-limit vacuum representation, not uniqueness among all Gibbs states or external-charge superselection sectors. It permits the vacuum complement to be zero.

6 Wilson normalization and a strong-coupling check

For pure four-dimensional $SU(2)$ Wilson theory, use normalized Haar per positively oriented link and one plaquette per coordinate pair:

$$Z_\beta(x) = \int \exp\left(\frac{\beta}{2}\sum_p \operatorname{Tr} U_p\right)\prod_e dU_e, \qquad \beta \ge 0. \tag{13}$$

Here $\operatorname{Tr} U_p$ is real and $|\operatorname{Tr} U_p/2| \le 1$. The convention $\beta \sum_p (1 - \operatorname{Tr} U_p/2)$ differs by an extensive factor that cancels in D. The structural transfer interface is recorded in `audits/su2_transfer_interface.md`; it must establish the exact normalized trace, gauge projection, operator positivity, and local insertions.

Once that interface is supplied, there is a crude direct strong-coupling check. With all periods at least four, every plaquette has four distinct link variables and Haar mean trace zero. There are $N_p(x) = 6 \prod_i x_i$ plaquettes. Jensen's inequality and a pointwise bound give

$$0 \le \log Z_\beta(x) \le 6\beta \prod_i x_i, \qquad D_\beta(4) \le 24576\beta.$$

For $0 \le \beta \le 2^{-26}$ the last bound is $3/8192 < 1/2048$. The criterion therefore recovers a contraction at most $2^{-1/4}$ on dyadic spatial cubes in this deliberately tiny strong-coupling interval, provided the structural interface holds. This is a consistency check, not a claimed advance in weak-coupling mass generation.

At $\beta = 0$, $Z = 1$, $D = 0$, and the physical transfer operator is rank one. The finite-slice gauge-invariant L^2 space is still infinite dimensional, but all its orthogonal-to-vacuum directions are killed by transfer. The reflected observable Hilbert space can therefore contain only the vacuum. Nontriviality is a genuinely separate requirement.

7 The exact missing continuum input

An intended application would choose an asymptotically free bare-coupling trajectory $\beta(a)$ and even $n(a)$ satisfying

$$an(a) \to R_0 \in (0, \infty), \qquad D_{\beta(a)}(n(a)) \le 1/2048 \quad \text{for all sufficiently small } a. \tag{14}$$

Then the cutoff transfer masses would satisfy

$$\frac{m_{\mathrm{lat}}(a)}{a} \ge \frac{\log 2}{an(a)} \longrightarrow \frac{\log 2}{R_0} > 0.$$

Equation (14) is *not proved here*. Even if it were, one would still need nontrivial limiting local Yang–Mills observables, the reconstruction axioms, the required short-distance behavior, and compatible convergence transferring the bound to their physical Hilbert space. Convergence of normalized semigroup matrix elements on a dense observable domain is one sufficient interface. A weak subsequential measure limit or a cutoff gap alone does not suffice; compare [3].

Proposition 7.1 (What a cutoff-comparison lemma would suffice to prove)**.** *Fix an integer blocking factor $B > 1$ and let $F_N(R) = \log Z_{\beta_N}(RB^N, RB^N, RB^N, RB^N)$ on admitted boxes. Suppose a common coupling trajectory β_N and volume-independent bulk constants c_N are chosen. Write $\widetilde{F}_N(R) = F_N(R) - c_N R^4$ and $\mathfrak{D}_N(R) = 16F_N(R) - F_N(2R)$. If, for one reference depth K, every $N \ge K$, and $R = M, 2M$,*

$$|\widetilde{F}_N(R) - \widetilde{F}_K(R)| \le CR^4 \epsilon_K,$$

then

$$|\mathfrak{D}_N(M) - \mathfrak{D}_K(M)| \le 32CM^4 \epsilon_K.$$

In particular, $\mathfrak{D}_K(M) \le \varepsilon_*/2$ and $32CM^4\epsilon_K \le \varepsilon_*/2$ imply a common successful physical box of side M at every finer cutoff. Under the structural transfer hypotheses, the cutoff physical transfer masses are at least $\log 2/M$.

Proof. The bulk terms cancel because $16M^4 = (2M)^4$. The triangle inequality bounds the two errors by $16CM^4\epsilon_K + C(2M)^4\epsilon_K = 32CM^4\epsilon_K$. Apply Theorem 1.2 with microscopic side $n_N = MB^N$ and spacing $a_N = B^{-N}$. □

Both inputs to this proposition remain unproved for the required Yang–Mills trajectory. The coupling and bulk subtraction must be common to M and $2M$; choosing a separate normalization for each box would invalidate the conclusion. The elementary strong-coupling check in Section 6 does not supply a matched weak-coupling reference at a depth where the comparison error is small.

The finite-box mechanism in [2] balances rough mixing against an error proportional to physical box volume. If a four-dimensional analogue supplied

$$\ell_{\mathrm{mix}}(K) \lesssim B^{\gamma K}, \qquad \mathrm{error}(K, R) \lesssim R^4 B^{-\alpha K}, \quad B > 1,$$

then $R = B^{\delta K}$ would be both large enough to mix and small enough to compare whenever $\gamma < \delta < \alpha/4$. Thus $4\gamma < \alpha$ is sufficient for *this particular strategy*, not necessary for Yang–Mills mass generation. Neither proposed Yang–Mills estimate is established here. Mixing must control the finite-volume defect or a sufficiently strong full-observable substitute; one selected two-point correlation is not automatically enough.

8 Verification and remaining scope

- The Python check verifies exact integer coefficient identities for all 24 telescoping orders and the rectangle identities, exact rational scalar bounds, and high-precision regressions on positive spectra and actual four-dimensional product-chain models. Examples do not replace proof.
- The formal directory records the exact scope of any Lean certificate. Rational constant certificates are not formalizations of the operator theorem or of Yang–Mills theory.
- Separate audits cover the source proof, the $SU(2)$ interface, adversarial examples, cutoff obligations, and prior literature.
- No published manuscript is modified. No claim is made that the full Family 215 continuum construction has been independently verified, that this criterion is new, or that the Millennium problem has been resolved.

References

[1] OpenAI, *Sharp mass bounds for the two-dimensional $O(4)$ model*, September 23, 2026. Supplied repository, section "A finite-volume criterion for the full gap," source `build/sections/finite-volume.tex`.

[2] OpenAI, *The canonical massive continuum limit of the two-dimensional $O(3)$ model*, October 4, 2026. Supplied repository, section "Trace estimates uniform in the cutoff," source `build/sections/trace.tex`.

[3] A. Jaffe and E. Witten, *Quantum Yang–Mills Theory*, in *The Millennium Prize Problems*, Clay Mathematics Institute and AMS, 2006, pp. 129–152. Official description, https://www.claymath.org/wp-content/uploads/2022/06/yangmills.pdf.